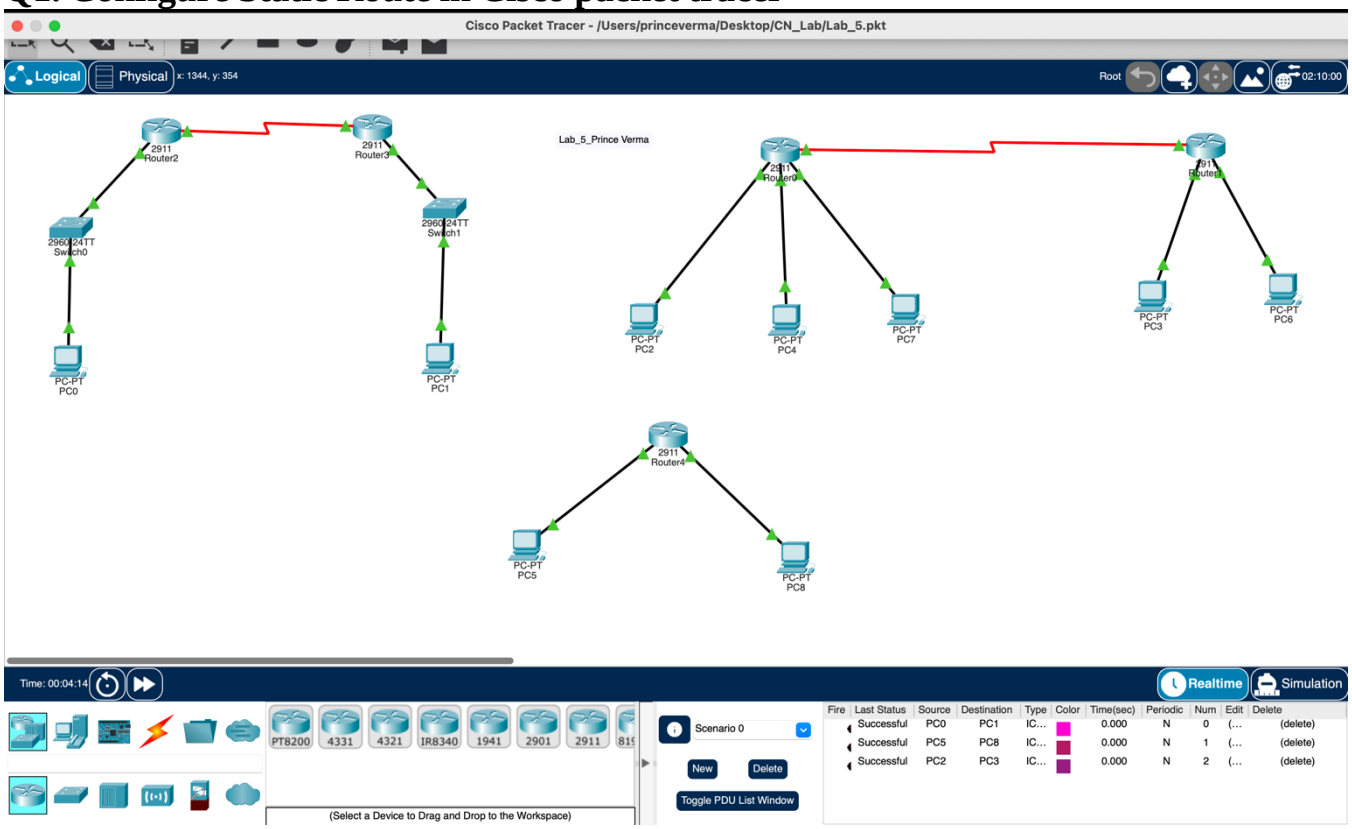
CN ASSIGNMENT 5

Q1. Configure Static Route in Cisco packet tracer



Commands for Router 1:

R1> enable
R1# configure terminal
R1(config)# interface gig 0/0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gig 0/0/1
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
R1(config-if)# exit

Commands for Router 2:

R2> enable

R2# configure terminal

R2(config)# interface gig 0/0/0

R2(config-if)# ip address 10.0.0.2 255.255.255.252

R2(config-if)# no shutdown

R2(config-if)# exit

R2(config)# interface gig 0/0/1

R2(config-if)# ip address 172.16.0.1 255.255.0.0

R2(config-if)# no shutdown

R2(config-if)# exit

Verify interface configuration

show ip interface brief

Display routing table before static routing

show ip route

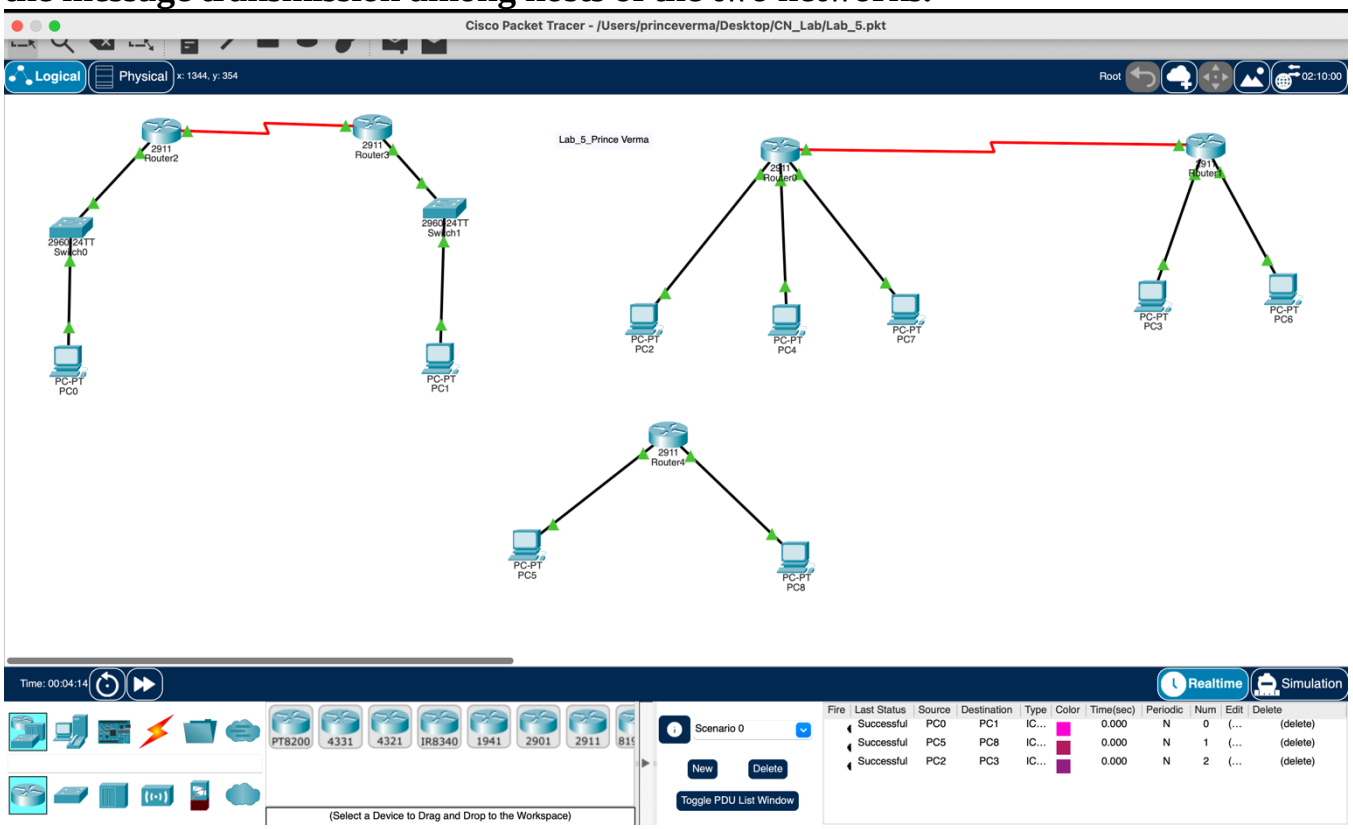
Configure static route on Router R1

R1(config)# ip route 172.16.0.0 255.255.0.0 10.0.0.2

Configure static route on Router R2

R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1

Q2. Suppose you want to create 3 subnets of a network having network address 198.168.1.0. In another network with network address 220.168.2.0, you want to create 2 subnets. For this scenario configure the end devices and routers accordingly. Connect two different networks (both are subnetted) through static routing and show the message transmission among hosts of the two networks.



Commands for Router 1

enable
configure terminal
interface gig0/0/0
ip address 198.168.1.1 255.255.255.192
no shutdown
interface gig0/0/1
ip address 198.168.1.65 255.255.255.192
no shutdown

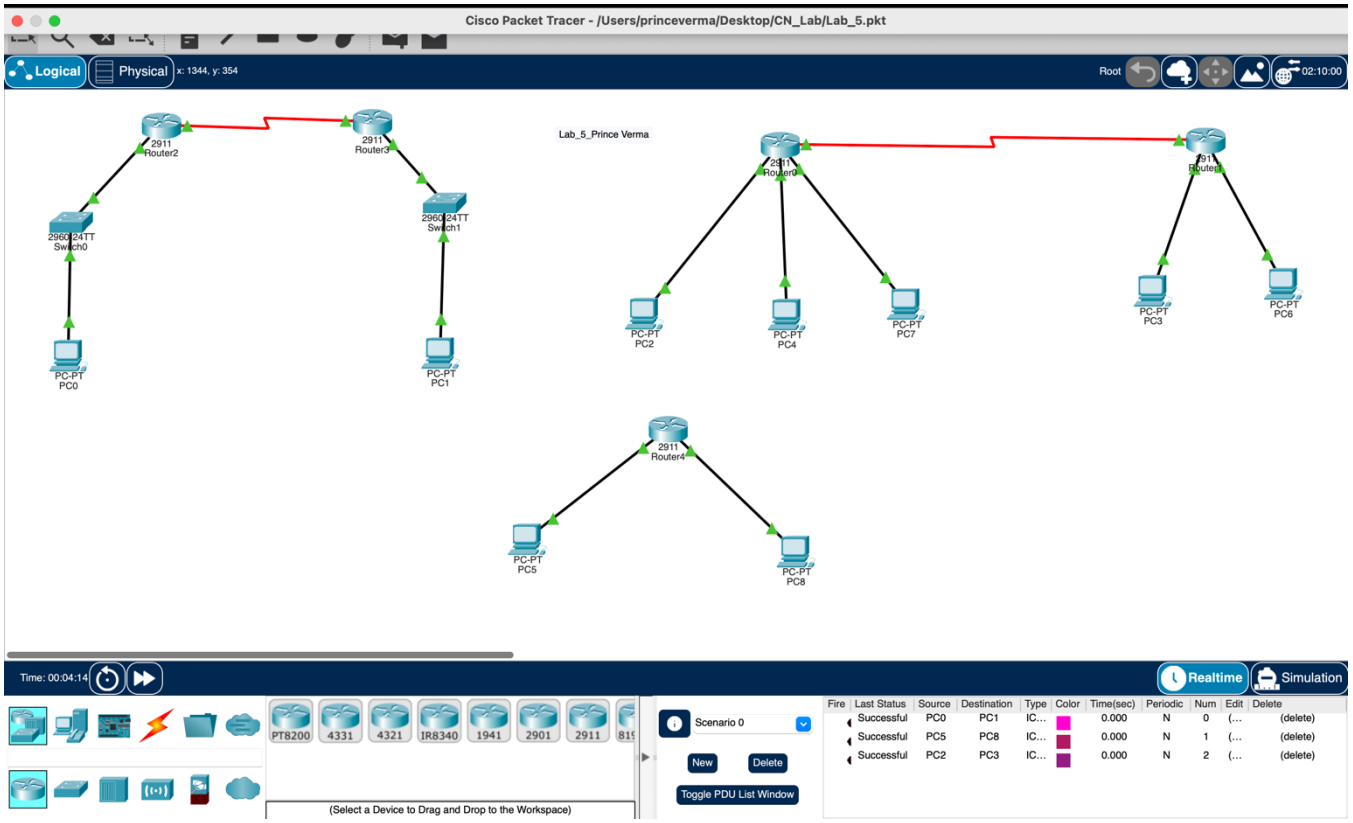
```
interface gig0/0/2
ip address 198.168.1.129 255.255.255.192
no shutdown
interface serial0/0/0
ip address 10.0.0.1 255.255.255.252
clock rate 64000
no shutdown
exit


ip route 220.168.2.0 255.255.255.128 10.0.0.2
ip route 220.168.2.128 255.255.255.128 10.0.0.2
end
```

Commands for Router 2

```
enable
configure terminal
interface gig0/0/0
ip address 220.168.2.1 255.255.255.128
no shutdown
interface gig0/0/1
ip address 220.168.2.129 255.255.255.128
no shutdown
interface serial0/0/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit
ip route 198.168.1.0 255.255.255.192 10.0.0.1
ip route 198.168.1.64 255.255.255.192 10.0.0.1
ip route 198.168.1.128 255.255.255.192 10.0.0.1
end
```

Q3. Connect two networks having network address 192.168.3.0 and 224.152.1.0 respectively. Network 1 having 4 and network two having 3 hosts. Connect two different networks through a router. For this scenario configure the end devices and routers accordingly and show the message transmission among hosts of the two networks.



Command for Router 1

enable
configure terminal
interface gig0/0/0
ip address 192.168.3.1 255.255.255.248
no shutdown
interface gig0/0/1
ip address 224.152.1.1 255.255.255.248
no shutdown
end